Real-sampled Biplex FFT (demuxed by 2)

Block: Real-sampled Biplex FFT (with output demuxed by 2) (`fft_biplex_real_2x`)
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Summary

Computes the real-sampled Fast Fourier Transform using the standard Hermitian conjugation trick to use a complex core to transform a two real streams. Thus, a biplex core (which can do 2 complex FFTs) can transform 4 real streams. Twiddle factor, and other logic sharing, allows multiples of 4 input streams to be processed simultaneously with minimal resource increases. Only positive frequencies are output (negative frequencies are the mirror images of their positive counterparts). Data is output in normal frequency order, meaning that channel 0 (corresponding to DC) is output first, followed by channel 1, on up to channel $2^{N-1} - 1$. Real inputs 0 and 2 share one output port (with the data for 0 coming first, then the data for 2), likewise for inputs 1 and 3, and so on.

Please note that this documentation refers to the latest version of this block and may not be valid for older versions, please look in the history for older versions of this documentation.

Mask Parameters

Parameter	Variable
Number simultaneous inputs (4*?)	n_inputs
Size of FFT: (2^?)	FFTSize
Input bit width	input_bit_width
Input binary point	

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Parameter	Variable
Asynchronous operation	async
Quantization Behavior	quantization
Overflow Behavior	overflow
Add Latency	add_latency
Mult Latency	mult_latency
BRAM Latency	bram_latency
Convert Latency	conv_latency
Number bits above which to store stage's coefficients in BRAM (2^? bits)	coeffs_bit_limit
Number bits above which to implement stage's delays in BRAM (2^? bits)	delays_bit_limit
BRAM sharing in coeff storage	coeff_sharing
Store a fraction of coeff factors where useful	coeff_decimation
Maximum fanout	max_fanout
Multiplier specification (0=core, 1=embedded, 2=behavioural) (left=1st stage)	mult_spec
Bit growth instead of shifting	bit_growth
Max bits to growth to	max_bits
Hardcode shift schedule	hardcode_shifts
Shift schedule	shift_schedule
DSP48 adders in butterfly	dsp48_adders

Ports

Port	Dir	Data Type	Descrip
sync	in	Boolean	sync is u
shift	in	Unsigned	Sets the
pol	in	Signed consisting of one (Input Bit Width) width signals per input.	The tim
en	in	Boolean	When a
sync_out	out	Boolean	Indicate
pol_out	out	Inherited	The frec
of	out	Unsigned, one bit per 4 inputs	Indicatic
dvalid	out	Boolean	Indicate

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